Practical Session 09: Financial NLP and Sentiment

Hands-On — LM Scoring, FinBERT vs. Lexicon, and a Filing-Date Event Study

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Session Overview

Duration: 1 hour (50 min work + 10 min debrief)

Agenda:

1. **Recap** (5 min): LM net sentiment, macro-F1, CAR.
2. **Problem 1** (12 min): Score an MD&A excerpt with the LM lexicon by hand.
3. **Problem 2** (12 min): Macro-F1 vs. accuracy on an imbalanced 3-class confusion matrix.
4. **Problem 3** (8 min): Build a $\mathrm{CAR}[1, 60]$ predictive regression.
5. **Case Study** (15 min): Run the chapter-09 companion notebook — LM lexicon figure + net sentiment.
6. **Discussion** (8 min): Lexicon vs. FinBERT vs. LLM in production.

Code

`code/notebooks/09-financial-nlp-sentiment/` — `demo.ipynb` (deterministic) and `exercises.ipynb` (live EDGAR lab).

Recap: The Three Formulas You Need

LM net sentiment (Problem 1):

$$S_{\mathrm{LM}}(d) = \frac{N_{\mathrm{pos}}(d) - N_{\mathrm{neg}}(d)}{N_{\mathrm{words}}(d)}$$

Macro-F1 over 3 classes (Problem 2):

$$F1_c = \frac{2\,\mathrm{P}_c \mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c}, \qquad \text{macro-F1} = \tfrac{1}{3} \sum_c F1_c$$

with $\mathrm{P}_c = \frac{TP_c}{TP_c + FP_c}$, $\mathrm{R}_c = \frac{TP_c}{TP_c + FN_c}$.

Cumulative abnormal return (Problem 3):

$$\mathrm{CAR}_i[1,60] = \sum_{\tau=1}^{60} \big(R_{i,\tau} - \hat{R}_{i,\tau}\big), \quad \mathrm{CAR}_i[1,60] = \alpha + \beta_1 \mathrm{SUE}_{i,0} + \beta_2 \tilde{S}_{i,0} + \varepsilon_i$$

Recap: Standardize Before You Compare

Raw scores are not comparable across firms or time. Z-score within industry-year:

$$\tilde{S}_{i,t} = \frac{S_{i,t} - \mu_{j(i),t}}{\sigma_{j(i),t}}$$

Why it matters for the regression

A defence or chemicals firm has structurally negative LM language. Without the $\tilde{S}$ transform, β_2 picks up an industry fixed effect, not a sentiment signal. The event study uses $\tilde{S}$, never the raw S_{LM}.

Also check: ADF/KPSS for long series; one fixed model vintage to avoid sentiment inflation; align timestamps to market hours (no look-ahead).

Problem 1: LM Scoring by Hand

Excerpt (10-K MD&A):

> *"Operating margins contracted as input costs rose. Management remains uncertain about demand but is confident the restructuring will deliver improved profitability."*

Tasks. Using the LM lists (negative dominates; *uncertain* is on the uncertainty list; *cost* is neutral in finance, not negative):

1. Identify LM negative, positive, and uncertainty hits.
2. Take $N_{\text{words}} = 24$. Compute S_{LM}.
3. A Harvard-GI classifier would also flag *costs* and *contracted*. Recompute the GI-style score. How large is the distortion?

Discussion: which word does the lexicon get *wrong* on negation, and how would FinBERT handle it?

Solution: Problem 1

LM hits:

- Negative: *contracted*, *uncertain*(also uncertainty) $\rightarrow$ count *contracted* as 1 negative.
- Positive: *confident*, *improved* $= 2$.
- Uncertainty: *uncertain* $= 1$ (hedging signal, not polarity).
- *costs*, *rose* are **neutral** in LM.

$$S_{\text{LM}} = \frac{2-1}{24} = +0.042 \quad \text{(mildly positive)}$$

GI-style: adds *costs* and *contracted* as negative $\rightarrow$ negatives $= 3$:

$$S_{\text{GI}} = \frac{2-3}{24} = -0.042$$

Sign flips — the GI false positives on neutral finance vocabulary reverse the conclusion. This is exactly the Loughran–McDonald (2011) critique.

Negation: "not confident" would still score *positive* under any lexicon; FinBERT's contextual [CLS] embedding resolves it.

Problem 2: Macro-F1 vs. Accuracy

A 3-class classifier on 1,000 financial sentences (neutral-heavy). Confusion matrix (rows = true, cols = predicted):

	pred neg	pred neu	pred pos	total
true neg	60	35	5	100
true neu	20	560	20	600
true pos	5	90	205	300

Tasks.

1. Compute overall accuracy.
2. Compute per-class precision, recall, F1 for **negative**.
3. Given $F1_{\text{neu}} \approx 0.86$ and $F1_{\text{pos}} \approx 0.74$, compute macro-F1.
4. Why is the gap between accuracy and macro-F1 the point of this exercise?

Solution: Problem 2

Accuracy $= (60 + 560 + 205)/1000 = \mathbf{0.825}$.

Negative class:

- $TP = 60$, $FP = 20 + 5 = 25$, $FN = 35 + 5 = 40$.
- Precision $= 60/85 = 0.706$; Recall $= 60/100 = 0.600$.
- $F1_{\text{neg}} = \dfrac{2(0.706)(0.600)}{0.706 + 0.600} = \mathbf{0.649}$.

Macro-F1 $= \frac{1}{3}(0.649 + 0.86 + 0.74) = \mathbf{0.750}$.

The lesson

Accuracy 0.825 looks strong, but it is propped up by the 600 easy neutral cases. Macro-F1 0.750 reveals the model is markedly weaker on the economically important **negative** class ($F1 = 0.65$). For a trading or risk signal, the neg/pos tails are what you act on — so report macro-F1.

Problem 3: Designing the Event Study

Setup. 800 earnings calls. For each you have: the call-day standardized text sentiment $\tilde{S}_{i,0}$, the earnings surprise $\mathrm{SUE}_{i,0} = (\mathrm{EPS} - \widehat{\mathrm{EPS}})/\hat{\sigma}_i$, and daily returns.

Tasks.

1. Write $\mathrm{CAR}_i[1, 60]$ in terms of abnormal returns. What benchmark $\hat{R}$ would you use, and why not raw returns?
2. State the regression and the null/alternative on β_2.
3. You find $\hat{\beta}_2 = 0.018$ ($t = 2.6$) after controlling for SUE. Interpret economically.
4. Name two robustness checks before you trust this.

Solution: Problem 3

1. $\mathrm{CAR}_i[1,60] = \sum_{\tau=1}^{60}(R_{i,\tau} - \hat{R}_{i,\tau})$, $\hat{R}$ from a market model / CAPM / factor benchmark. Raw returns mix in market-wide moves; abnormal returns isolate the firm-specific drift.

2. $\mathrm{CAR}_i[1,60] = \alpha + \beta_1 \mathrm{SUE}_{i,0} + \beta_2 \tilde{S}_{i,0} + \varepsilon_i$. $H_0 : \beta_2 = 0$ (text adds nothing beyond the EPS number); $H_1 : \beta_2 > 0$ (residual text sentiment predicts drift).

3. $\hat{\beta}_2 = 0.018$, $t = 2.6$: a one-SD increase in standardized sentiment $\rightarrow$ $+1.8$ pp of 60-day abnormal return, *orthogonal* to the quantitative surprise. Text carries incremental information — post-announcement drift conditioned on tone.

4. Robustness: (i) strictly out-of-sample period not used for model selection (economic-validity test); (ii) cluster SEs by firm/date; also check look-ahead (timestamp vs. market close) and a placebo pre-event window.

Case Study: Chapter-09 Companion Notebook

Open `code/notebooks/09-financial-nlp-sentiment/demo.ipynb`. It is deterministic (no network) and reproduces the LM lexicon figure + the worked net-sentiment example.

```python
import matplotlib.pyplot as plt

# Loughran-McDonald finance dictionary word-list sizes.
lists = {"Negative": 2355, "Litigious": 903, "Positive":
    354,
         "Uncertainty": 297, "Weak modal": 27, "Strong modal
             ": 19}
items = sorted(lists.items(), key=lambda kv: kv[1])
plt.barh([k for k, _ in items], [v for _, v in items], color
    ="#4c72b0")
plt.xlabel("Number of words"); plt.title("Loughran-McDonald
    dictionary")
plt.tight_layout(); plt.show()
```

Case Study: Net LM Sentiment Example

The same notebook reproduces the worked net-sentiment number from the chapter.

```
# Net LM sentiment for the chapter MD&A excerpt (ex:lm-
    example).
n_pos, n_neg, n_words = 1, 4, 35
s_lm = (n_pos - n_neg) / n_words
print(f"S_LM = ({n_pos} - {n_neg}) / {n_words} = {s_lm:.3f}"
    )  # ~ -0.086
```

Case Study: Diagnostic Questions

Run `demo.ipynb`, then answer:

Q1. The negative list (2,355) dwarfs the positive list (354). What does this asymmetry tell you about the language of corporate disclosure — and about the *base rate* a lexicon will produce on a typical 10-K?

Q2. The excerpt scores $S_{\mathrm{LM}} = -0.086$. Re-run with `n_pos, n_neg = 2, 2`. How sensitive is the sign to a single word-list membership decision? What does this imply for reproducibility?

Q3 (live lab). Move to `exercises.ipynb`: it pulls real 10-Ks from EDGAR and prices from `yfinance`. Score one filing, then test whether filing-day LM negativity is associated with the filing-day abnormal return. Does the sign match Loughran–McDonald (2011)?

Extension

Swap the lexicon for `yiyanghkust/finbert-tone` on the same filings. Where do the two methods *disagree*, and which sentences drive it?

Discussion: Choosing a Method in Production

For each scenario, pick lexicon / FinBERT / LLM and justify on cost, latency, labels, interpretability, and accuracy.

1. A regulator must audit *why* every sentiment score on a 10-K was assigned. 50k filings/year, no labelled data, full transparency required.
2. A desk scores 5M news headlines/day with a < 50 ms/headline budget and 10k labelled examples in hand.
3. A research team prototypes sentiment on a brand-new document type (crypto whitepapers) with *zero* labels and a small corpus.

Frame your answer around

Auditability (lexicon wins), throughput/latency (distilled FinBERT wins), cold-start flexibility (LLM zero-shot wins) — and when you would *distil* an LLM into a small model for scale.

Wrap-Up: Key Takeaways

- **Lexicons are auditable but brittle:** the GI vs. LM sign flip in Problem 1 is the whole case for domain-specific dictionaries.
- **Report macro-F1, not accuracy:** the neutral majority hides weakness on the actionable tails (Problem 2).
- **Standardize, then test economically:** z-score within industry-year, run the CAR regression, validate strictly out-of-sample (Problem 3).
- **Method choice is a trade-off:** auditability vs. throughput vs. cold-start flexibility — no universal winner.

Next practical

Practical 10 — domain-specific LLMs: fine-tuning and adaptation of Fin-BERT's successors on long financial documents.

Appendix

Stretch Problems

Extra / optional material — beyond the core lecture.

Stretch: Krippendorff's Alpha

Two annotators label 5 sentences (P=pos, N=neu, G=neg):

	s1	s2	s3	s4	s5
Annot. A	P	N	G	N	P
Annot. B	P	N	G	G	P

Tasks. (1) Compute observed disagreement D_o (fraction of disagreeing pairs). (2) With marginal frequencies, compute $D_e = 1 - \sum_c (n_c/n)^2$. (3) Compute $\alpha = 1 - D_o/D_e$. Is it above Krippendorff's 0.667 reliability bar? (4) Why would the *ordinal* variant give a different, more lenient verdict on the s4 P/G... here N/G disagreement?

Stretch: From Score to Sharpe

Setup. A standardized news-sentiment signal $\tilde{S}$ is rebalanced daily into a long–short portfolio (long top decile, short bottom decile).

Tasks.

1. The in-sample annualized Sharpe is 1.4; strictly out-of-sample it is 0.5. Name three reasons for the decay (hint: one is in the lecture's "sentiment inflation" box).
2. Kirtac & Germano (2024) report OPT beats the LM dictionary on long–short Sharpe. Why might the gap shrink *after* transaction costs at high turnover?
3. Lehner (2024) shows LLMs can rewrite filings to shift measured tone. How does adversarial text-generation threaten a filings-based signal's out-of-sample stability?